

Yuhang Zhou

email: zyhang@meta.com
734-882-8851

EMPLOYMENT

Meta, New York, the U.S.

▪ *Research Scientist*

May 2025-

- Pioneered the design of a large language model tailored for next-generation recommendation systems, enhancing personalization and relevance.

EDUCATION

University of Maryland, College Park, the U.S.

Ph.D. in Information Studies

May 2025

- Coursework: Natural Language Understanding, Computational Linguistics

GPA: 3.98/4.0

University of Michigan, Ann Arbor, the U.S.

Bachelor of Science in Data Science

May 2020

- Coursework: Deep Learning for Vision, Data Mining, Database Management System

GPA: 3.90/4.0

Shanghai Jiao Tong University, Shanghai, China

Bachelor of Science in Electrical and Computer Engineering

Aug 2020

- Coursework: Data Mining and Machine Learning, Honors Mathematics, Discrete Mathematics

GPA: 3.56/4.0

RESEARCH

University of Maryland

▪ *Concept Spurious Correlation in LLMs, advised by Prof. Furong Huang*

- Explored the spurious correlation at the concept level for large language model fine-tuning or in-context learning.
- Utilized the advanced LLMs to generate the “counterfactual” sentences to resolve the spurious correlation.
- Data augmentation method can bring 1.44% relative improvement on classification benchmarks and effectively mitigate performance bias.

▪ *LLM Long-tailed Knowledge Distillation, advised by Prof. Furong Huang*

- Developed a framework with active learning and data augmentation to resolve the challenges of LLM knowledge distillation on long-tailed datasets.
- Proposed framework can bring about a relative improvement of 6.81% on long-tailed datasets.

University of Maryland, CLIP Lab

▪ *Token Diffusion on the Social Networks, advised by Prof. Wei Ai*

- Explored the diffusion patterns of newly created emojis on social media on the frequency and semantic aspects.
- Discovered the negative effect of new emojis on the sentiment classification task of language models.
- Proposed a emoji substitution method to resolve out-of-vocabulary challenges and lead to a 6.92% relative improvement on tweets with new emojis.

SELECTED PUBLICATION

* denotes equal contribution. See full publications in Google Scholar

- Nuoya Xiong, **Zhou, Yuhang**, Hanqing Zeng, Zhaorun Chen, Furong Huang, Shuchao Bi, Lizhu Zhang, and Zhuokai Zhao. Token-level llm collaboration via fusionroute. *ICML 2026*, 2026
- **Zhou, Yuhang**, Mingrui Zhang, Ke Li, Mingyi Wang, Qiao Liu, Qifei Wang, Jiayi Liu, Fei Liu, Serena Li, Weiwei Li, et al. Mixture-of-minds: Multi-agent reinforcement learning for table understanding. *ACL 2026*, 2025a
- **Zhou, Yuhang**, Jing Zhu, Shengyi Qian, Zhuokai Zhao, Xiyao Wang, Xiaoyu Liu, Ming Li, Paiheng Xu, Wei Ai, and Furong Huang. Disco balances the scales: Adaptive domain-and difficulty-aware reinforcement learning on imbalanced data. *Findings of EMNLP 2025*, 2025b
- **Zhou, Yuhang**, Giannis Karamanolakis, Victor Soto, Anna Rumshisky, Mayank Kulkarni, Furong Huang, Wei Ai, and Jianhua Lu. Mergeme: Model merging techniques for homogeneous and heterogeneous moes. *NAACL 2025*, 2025c
- **Zhou, Yuhang***, Jing Zhu*, Paiheng Xu, Xiaoyu Liu, Xiyao Wang, Danai Koutra, Wei Ai, and Furong Huang. Multi-stage balanced distillation: Addressing long-tail challenges in sequence-level knowledge distillation. *Findings of EMNLP 2024*, 2024
- **Zhou, Yuhang**, Paiheng Xu, Xiaoyu Liu, Bang An, Wei Ai, and Furong Huang. Explore spurious correlations at the concept level in language models for text classification. *ACL 2024*, 2024
- **Zhou, Yuhang** and Wei Ai. Teaching-assistant-in-the-loop: Improving knowledge distillation from imperfect teacher models in low-budget scenarios. *Findings of ACL 2024*, 2024

- Xiyao Wang, **Zhou, Yuhang**, Xiaoyu Liu, Hongjin Lu, Yuancheng Xu, Feihong He, Jaehong Yoon, Taixi Lu, Gedas Bertasius, Mohit Bansal, et al. Mementos: A comprehensive benchmark for multimodal large language model reasoning over image sequences. *ACL 2024*, 2024
- Jing Zhu*, **Zhou, Yuhang***, Vassilis N Ioannidis, Shengyi Qian, Wei Ai, Xiang Song, and Danai Koutra. Pitfalls in link prediction with graph neural networks: Understanding the impact of target-link inclusion & better practices. In *ACM International Conference on Web Search and Data Mining (WSDM)*, 2024

INTERNSHIP

Amazon AGI , New York, USA

- *Applied Scientist Intern, Manager: Giannis Karamanolakis* May 2024-Aug 2024
 - Designed a novel MoE (Mixture of Experts) architecture to incorporate heterogeneous expert models.
 - Proposed multiple routing heuristics to improve MoE performance when merging without fine-tuning.

Amazon Alexa AI, Boston, USA

- *Applied Scientist Intern, Manager: Yuguang Yue* May 2023-Aug 2023
 - Designed a unified large language model knowledge distillation framework with student model signals to improve the student model fine-tuning performance.
 - Conducted extensive experiments on datasets and multiple public models with the corresponding promoting methods.
 - Proposed two-stage framework brings a relative improvement of up to 20.79% compared to random fine-tuning (paper published in *Findings of ACL*, 2024)

Amazon Alexa AI, New York, USA

- *Applied Scientist Intern, Manager: Beiye Liu* May 2022-Aug 2022
 - Designed a new prompt-tuning semi-supervised learning pipeline for large language models without manual prompts and verbalizers to promote the scalable text classification.
 - Reproduced the state-of-the-art few-shot prompt-tuning methods as the experiment baseline results.
 - Proposed method brings a relative improvement of 2.52% over the previous method with manual prompts and verbalizers (paper published in *Findings of EACL*, 2023)

Alibaba Group, Hangzhou, China

- *Research Intern, Manager: Tianyu Li* May 2021-Aug 2021
 - Applied Siamese BERT-Networks to retrieve items of Taobao, given particular style descriptions.

SKILLS

- Language: Proficient in Python, SQL, $\LaTeX$, C++, C, R, Java, JavaScript